

Research paper

Silicon weakens the outer apoplastic barrier in roots of rice and delays its formation, resulting in increased Na⁺ and Cl⁻ fluxes to the shootShuai Tong^{a,*}, Kosala Ranathunge^b, Elisa Pellegrini^c, Dan Liu^a, Max Herzog^a, Juan de la Cruz Jiménez^a, Lukasz Kotula^{d,1}, Ole Pedersen^{a,b,**}^a Department of Biology, University of Copenhagen, Universitetsparken 4, 3rd floor, Copenhagen 2100, Denmark^b School of Biological Sciences, The University of Western Australia, 35 Stirling Highway, Perth, WA 6009, Australia^c Department of Agricultural, Food, Environmental and Animal Sciences, University of Udine, Via delle Scienze 206, Udine 33100, Italy^d UWA Institute of Agriculture, The University of Western Australia, 35 Stirling Highway, Perth, WA 6009, Australia

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ABSTRACT

In rice, silicon can mitigate abiotic and biotic stresses. We therefore investigated the effect of Si on key root traits related to soil flooding and salinity tolerance with emphasis on the outer apoplastic barrier and cortical aerenchyma. We tested the hypothesis that Si application alters the phenotypic response of these root traits by growing rice in nutrient solutions without or with Si, designed to mimic drained or flooded soils. We measured the barrier strength through resistance to O₂ and water of the outer parts of adventitious roots along with cortical aerenchyma and other root structural traits. We found that Si delayed the barrier formation and caused lower amounts of inducible cortical aerenchyma. The delay in barrier formation resulted in higher xylem loading of Na⁺ and Cl⁻, i.e., the sap flux of both ions was significantly higher for plants with access to Si. The increased ion fluxes correlated with lower lignin and suberin deposition in the outer part of the root. Consequently, we do not recommend using Si application to alleviate combined stress of salinity and soil flooding in rice, since the barrier was more permeable to O₂, and the aerenchyma formation was less pronounced in roots with Si.

1. Introduction

Silicon (Si) is known for its beneficial effects on rice in a range of environmental conditions. Although Si has not traditionally been considered an essential element for rice growth, it does provide significant benefits by mitigating various forms of stress (Debona et al., 2017; Epstein, 2009; Thakral et al., 2021), but its possible effect and mode of action is being questioned (Coskun et al., 2019). Positive effects on abiotic stress alleviation include studies on salinity (Gong et al., 2006), heavy metals (Feng Shao et al., 2017; Xiao et al., 2022), and flooding where Si application can shape root morphological traits and increase plant biomass accumulation (Pan et al., 2021). Over 80 % of rice production worldwide is from lowland rice grown on flooded soils, and rainfed rice can also be severely impacted by soil flooding or even submergence (Nishiuchi et al., 2012). However, understanding the effects and mechanisms of Si in alleviating combined stress of salinity and soil flooding in rice remains very limited.

Silicon can modify root apoplastic barriers, which are involved in tolerance to soil flooding. Previous research has demonstrated that Si can enhance suberization and lignification in cell walls of the exodermis of the adventitious roots of rice, leading to a notable reduction in radial O₂ loss via modification of the outer apoplastic barriers (Fleck et al., 2011; Hinrichs et al., 2017). However, the impact of Si on the development and configuration of apoplastic barriers remains a subject of debate (Fleck et al., 2015; Hinrichs et al., 2017; Kreszies et al., 2020). Several studies exploring the influence of Si on suberized barriers in roots have primarily focused on the interplay between Si and various abiotic stresses in plants, such as osmotic stress, with results reporting instances of enhanced, unaffected, or even delayed barrier formation (Kreszies et al., 2020).

Interestingly, the deposition of Si in the exodermal and endodermal cell walls can reduce the apoplastic bypass flow of sodium (Na⁺), subsequently reducing the Na⁺ accumulation in shoots (Flam-Shepherd et al., 2018; Gong et al., 2006; Yeo et al., 1999). The mechanism behind

* Corresponding author.

** Corresponding author at: Department of Biology, University of Copenhagen, Universitetsparken 4, 3rd floor, Copenhagen 2100, Denmark
E-mail addresses: shuai.tong@bio.ku.dk (S. Tong), ole.pedersen@uwa.edu.au (O. Pedersen).¹ Current address: Department of Primary Industries and Regional Development, 3 Baron-Hay Court, Perth, WA 6151, Australia

this reduction is not fully understood, but it is likely attributed to the polymerization of silicates in the endodermis and exodermis of adventitious roots (Flowers et al., 2010; Maathuis et al., 2014). However, limited information is available regarding the potential effect of Si on the bi-directional apoplastic flux of Na^+ within rice roots (Malagoli et al., 2008), and given the established ability of Si to reduce apoplastic bypass flow in rice (Flam-Shepherd et al., 2018), it is worth investigating this potential relationship.

Outer apoplastic barriers and root tissue porosity (including aerenchyma and intercellular gas-filled spaces) both serve to deliver molecular O_2 for root respiration in anoxic soils. In studies focusing on radial O_2 loss, the outer apoplastic barrier is referred to as the barrier to radial O_2 loss (Peralta Ogorek et al., 2024). While certain wild rice species constitutively form a barrier to radial O_2 loss (Ejiri et al., 2020), it is induced in cultivated rice as a response to soil flooding (Colmer, 2003a). The barrier restricts O_2 loss through radial diffusion into the anaerobic rhizosphere due to the deposition of lignin and/or suberin in the cell walls of the outer root cell layers (Colmer, 2003b; Colmer et al., 2019; Pedersen et al., 2021). The barrier to radial O_2 loss serves not only to impede the diffusion of O_2 , but also acts as a barrier against other gases, such as hydrogen (H_2), water vapour and hydrogen sulphide (H_2S) (Peralta Ogorek et al., 2021, 2023). Moreover, the barrier to radial O_2 loss also protects the plant from soil phytotoxins including H_2S (Peralta Ogorek et al., 2023) and reduced iron (Fe^{2+}) (Jiménez et al., 2021a). Apoplastic barriers (in the endodermis as well as in the outer part of the root) significantly affect apoplastic solute movement, curbing excessive influx into roots, and it has been shown that solute permeability reduction was associated with apoplastic barriers in the exodermis (Hose et al., 2001). Moreover, there is a striking correlation between apoplastic barrier permeability and Na^+ and Cl^- transports from the root medium to the shoot (Krishnamurthy et al., 2011; Ranathunge et al., 2011; Ranathunge and Schreiber, 2011). These barriers can impede solute flow, reduce Na^+ uptake in shoots and increase resilience of rice under subsequent acute salt stress (Krishnamurthy et al., 2009, 2011). Overall, these findings propose that root apoplastic barriers restrict Na^+ bypass, ultimately enhancing plant viability during salinity stress.

Cortical aerenchyma is another root trait that enables rooting in flooded soils. The aerenchyma serves as a low-resistance pathway, facilitating longitudinal molecular diffusion of O_2 from the shoot to the root apex (Armstrong, 1979). The longitudinal O_2 supply is essential in flooded soils as these are normally anoxic preventing uptake of O_2 from the rhizosphere (Ponnamperuma, 1972). In rice, the root cortex always contains some amounts of constitutively formed aerenchyma present also in drained soils, but the amount (inducible aerenchyma) is significantly increased as a response to soil flooding (Yamauchi et al., 2013). Collectively, aerenchyma and the apoplastic barriers of the root peripheral cell layers, representing diverse morphological and anatomical features, are indispensable for growth in flooded soils.

Si application can benefit rice growth; however, little is known about the effect of Si on flood-related root traits in flooded, anoxic soils. Therefore, we aimed at evaluating the effects of Si application on roots of rice grown under contrasting O_2 conditions. We specifically evaluated how Si affects the barrier to radial O_2 loss (influencing root permeability to O_2 and Na^+ and Cl^- fluxes from roots to shoots) and aerenchyma formation. Based on already reported beneficial effects of Si application, we hypothesized that Si reduces the permeability of the barrier to radial O_2 loss, restricts the radial bypass of Na^+ and Cl^- at non-stress concentrations, and increases the inducible aerenchyma. We used a multi-disciplinary approach, diagnosing anatomical and physiological traits using, e.g., state-of-the-art O_2 microsenors and a novel lignification index. In this way, our study has filled a crucial knowledge gap by evaluating the significance of Si for key root traits contributing to soil flooding tolerance.

2. Materials and methods

2.1. Experimental design

Measurements were conducted in hydroponics under stagnant, deoxygenated growing conditions (mimicking soil flooding cf. Wiengweera et al. 1997) to investigate the formation of the barrier to radial O_2 loss, inducible aerenchyma, and other root anatomical traits known to enhance tolerance to soil flooding. We also measured Na^+ and Cl^- concentrations in the xylem sap following growth in stagnant, deoxygenated agar. The remaining root traits were also assessed in aerated and non-aerated nutrient solutions mimicking a normoxic (drained) and a hypoxic soil, respectively (Table 1); see Fig. S1 for an overview of the experimental design.

2.2. Plant material, growing conditions and treatments

We used a lowland rice (*Oryza sativa* L.) cultivar, IR42, known to be moderately drought tolerant (Ponnamperuma, 1979) and for being a submergence-susceptible check (Ismail et al., 2009). Moreover, IR42 has recently been used as a model cultivar for studies on outer apoplastic barriers in roots (Peralta Ogorek et al., 2024). It is known to form a barrier to radial O_2 loss when growing in stagnant, deoxygenated conditions and this barrier has recently been shown to also restrict radial water loss (Song et al., 2023) and to protect the root from phytotoxins such as H_2S (Peralta Ogorek et al., 2023). Seeds from the International Rice Research Institute (IRRI, Manila, Philippines), were surface sterilized in 70 % ethanol for 1 min, followed by immersion in 3.5 % NaClO for 30 min before being washed three times with deionized (DI) water (Becker et al., 2020). The seeds were then immersed in aerated 0.5 mM CaSO_4 for 3 h, moved to Petri dishes with tissue paper moistened with 0.5 mM CaSO_4 and placed at 28 °C in the dark for three days to germinate. The seedlings were placed on a mesh floating on aerated 25 % strength nutrient solution without Si (see below for full-strength nutrient solution) and grew for four days in a controlled temperature chamber at: 30/25 °C (day/night), a 14/10 h light/dark cycle with a photon flux of $\sim 500 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$, and a relative humidity exceeding 70 %. The seedlings were transplanted to 4 L plastic pots with perforated lids (eight plants per pot) containing 100 % strength nutrient solution. Seedlings (1-week-old) were then transplanted into 6-litre plastic pots with perforated lids (three plants per pot) containing

Table 1
Root traits investigated under contrasting growing conditions.

Trait	Aerated	Non-aerated	Stagnant
Visualization of O_2 loss from roots	✓	✓	✓
Permeability of the root apoplast to a tracer dye	✓	✓	✓
Volume of aerenchyma in roots	✓	✓	✓
Apparent permeance to O_2 of the outer cell layers of the root	✓	✓	✓
Radial water loss from roots	✓	✓	✓
Root anatomy	✓	✓	✓
Plant biomass	✓	✓	✓
Number of roots per plant	✓	✓	✓
Maximum root length	✓	✓	✓
Root extension rate	✓	✓	✓
Barrier against the radial O_2 loss along the root length			✓
Root porosity (including aerenchyma and intercellular gas-filled spaces)			✓
Xylem sap Na^+ and Cl^- concentrations			✓
Whole root surface area			✓
Root lignification			✓
Root suberization			✓

Notes: Aerated = aerated nutrient solution; Non-aerated = non-aerated nutrient solution; Stagnant = stagnant, deoxygenated nutrient solution.

full-strength nutrient solution. Plants were held individually in each of the three holes in the lids using sliced 20 mm polyethylene foam plugs, and the pots were covered with aluminium foil to exclude light from the root system. The composition of the full-strength nutrient solution was as follows (in mM): K^+ , 4; Ca^{2+} , 1.5; NH_4^+ , 0.625; Mg^{2+} , 0.4; Na^+ , 4.0;

NO_3^- , 4.4; SO_4^{2-} , 1.9; $H_2PO_4^-$, 0.2; SiO_3^{2-} , 0 (without Si) or 2 (with Si); Mn^{2+} , 0.002; Zn^{2+} , 0.002; Ni^{2+} , 0.001; Cu^{2+} , 0.0005; Cl^- , 4.05 (without Si) or 0.05 (with Si); BO_3^{3-} , 0.025; MoO_4^{2-} , 0.0005; Fe-EDTA, 0.05; MES buffer, 2.5 (pH adjusted to 6.0 using 10 % H_2SO_4 or 2 M KOH, with the K^+ and SO_4^{2-} concentrations excluded in the value stated above).

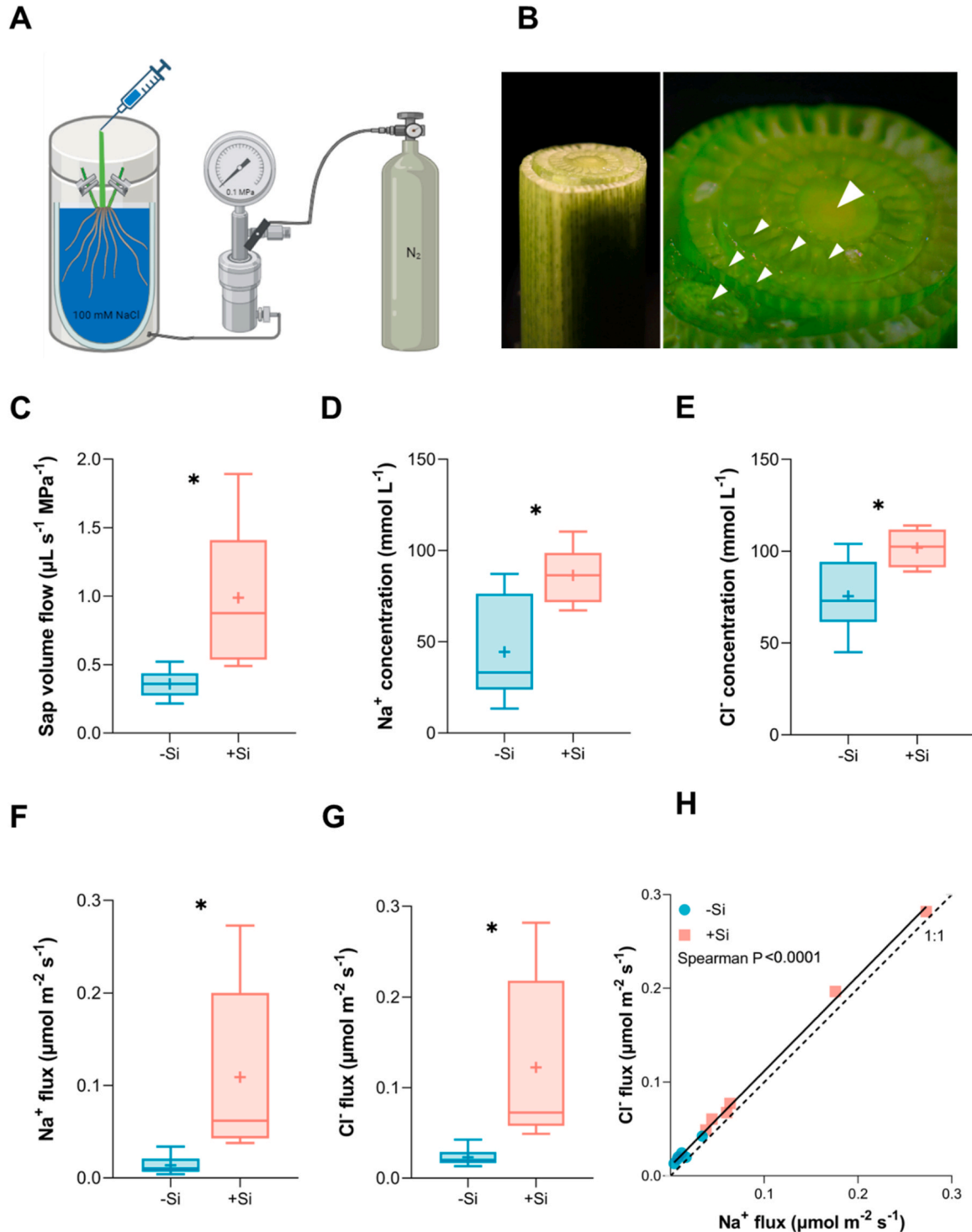


Fig. 1. Concentrations of Na^+ and Cl^- in the xylem sap of *Oryza sativa* grown in the stagnant medium without or with 2 mM Si application. (A) Experimental setup for collection of xylem sap from basal part of the stem for roots exposed to a 100 mM NaCl solution. Rice plants of 4–5-week-old were grown in stagnant, deoxygenated nutrient solution, without or with Si. The root system was pressurized to 0.1 MPa in Scholander chamber (A) and xylem sap was collected for 1 h from the main stem (B) was collected for 1 h to calculate the flux (C) with the white arrows indicating area xylem sap concentration. (D) and (E) show the concentrations of Na^+ and Cl^- , respectively. (F) and (G) show the fluxes of Na^+ and Cl^- , respectively and (H) the Spearman correlation between the two ions ($P < 0.0001$, $r = 0.979$, $n = 24$). The box and whisker plot show the mean (+) of six true replicates, the median (horizontal line), the first and third quartiles (box), and minimum and maximum (whiskers). Asterisks indicate significant differences (Unpaired, 2-tailed Students t -test): $*P < 0.05$; ns, not significant.

The experiment involved six treatments in nutrient solution: (i) aerated, no Si; (ii) aerated, with 2 mM Si; (iii) non-aerated, no Si; (iv) non-aerated, with 2 mM Si; (v) stagnant, no Si; and (vi) stagnant, with 2 mM Si. Silicon was supplied in the form of sodium silicate (Na_2SiO_3) in nutrient solution (in hydroponic studies of rice, Si supplementation typically ranges from 1.7 to 2.0 mM (Avila et al., 2019; Park et al., 2019) with uptake becoming saturated at 1.28 mM Si (Ma et al., 2001)). Moreover, the chosen Si concentration for Si treatment in the present study (2 mM) reflects typical soil water concentrations of Si in paddy soil (0.2–1.2 mM) (Mihara et al., 2017; Ronchi et al., 2018). In treatments without Si application, NaCl was added to compensate for the potential effect of Na^+ supplied by Na_2SiO_3 (Kreszies et al., 2020), but not to stress-inducing levels (4 mM). All chemical substances used were of analytical grade, and only plasticware was used to prevent Si contamination from glassware. There were five replicate pots for each treatment (30 pots in total), and the pots were randomized and considered the experimental unit (replicate). Nutrient solution was renewed weekly during the initial 3 weeks, followed by renewal every 3 d thereafter.

Aerated nutrient solution was continuously purged with air at a flow rate of $\sim 80 \text{ ml min}^{-1}$. No aeration was provided to non-aerated or to the stagnant, deoxygenated nutrient solution (hereafter stagnant medium). Stagnant medium contained 0.1 % (w/v) agar and was deoxygenated by purging with N_2 for 3 h. Agar prevents the convection and reintroduction of atmospheric gases into the root medium and simulates slow gas diffusion as it occurs in flooded soils (Wiengweera et al., 1997). Plants assigned to stagnant conditions received a hypoxic pre-treatment by purging the 25 % nutrient solution with N_2 for 5 min before these were left overnight without bubbling. The purpose of this pre-treatment was to prevent anoxic shock (Gibbs and Greenway, 2003).

When the rice seedlings were 3-week-old, the $p\text{O}_2$ (Opto-MR, Unisense, Denmark) and osmotic concentrations (3250: Advanced Instruments, USA) of the nutrient solutions were both measured 60 mm below the surface of the solution (Fig. S2A, B) every 24 h for 3 d. When the plants were 4-week-old, 10–30 mm long roots were marked with thread, and their root length was measured every 24 h for 3 d. Up until a length of 80–100 mm, the roots exhibited similar extension rates ($26\text{--}29 \text{ mm d}^{-1}$) across all treatments (Fig. S2C) ensuring that the target tissues were of similar age, and that all target roots had developed under the experimental conditions described above. The maximum root length and the number of roots were recorded from one plant in each pot after 4 weeks of treatments. Additionally, the shoot and root of one randomly selected plant from each pot was harvested and dried at 60°C for 4 d before being weighed and dry mass calculated.

2.3. Xylem sap collection

Xylem sap was collected from 4-week-old plants grown in stagnant medium with or without 2 mM Si following the procedure of Miyamoto et al. (2001), using a water status console (Model 3000F01, Soilmoisture Equipment Corp). At first, all tillers, except the main stem, were cut 30 mm from the base and closed using clamps, and the root-shoot junction was positioned parallel to the surface of a 100 mM NaCl solution. During measurements, the roots were enclosed in a Scholander chamber to apply pneumatic pressure to the root medium (Fig. 1A). The base of the main stem was tightly sealed with custom-made grommets, support washers and silicone seals pre-drilled with a hole matching the diameter of the stem. For fine adjustments, Parafilm and plastic wrap were used to wrap the main stem. To induce sap flow from roots to the cut surface of the main stem, the pressure within the pressure vessel was raised to 0.1 MPa by compressed N_2 above atmospheric pressure. The xylem sap that emerged from the cut surface of the main stem was collected over 1 h using a custom-made plastic capillary (Fig. 1B) and the sap was then transferred to Eppendorf tubes and stored for further analysis. After the sap collection, plants were harvested, and their roots were separated and scanned at 600 dpi using a flatbed scanner (Ricod, IM C300). The scanned root systems were subsequently analysed using

RhizoVision software to obtain root surface areas.

2.4. Determination of Na^+ and Cl^- concentrations in the xylem sap

Na^+ and Cl^- concentrations in the collected xylem sap (80–700 μl) were measured using a flame photometer (Flame Photometer PFP7) and a chloridometer (Gonotec Chloridimeter CM 20), respectively. Na^+ and Cl^- fluxes were calculated as follows:

$$F = \frac{C \bullet V}{S \bullet 3600} \quad (1)$$

where F is the flux ($\mu\text{mol m}^{-2} \text{ s}^{-1}$), C (mM) is the ion concentration, V (ml) is the volume of the collected sap (determined as the mass of the collected sap assuming a density of 1 g cm^{-3}), S (m^2) is the surface area of root system, and 3600 represents the conversion from h to s.

2.5. Visualization of radial O_2 loss from roots

Radial O_2 loss from adventitious roots were qualitatively visualized by methylene blue staining following the method described in (Jiménez et al., 2021b). Methylene blue is colourless in its reduced form but turns blue upon oxidation. One to three adventitious roots (90–140 mm long) from a 4-week-old plant were selected and all other roots were trimmed off just below the root-shoot junction. The plant was then moved to a transparent glass container ($350 \times 250 \times 15 \text{ mm}$) filled with deoxygenated 0.1 % agar solution and reduced methylene blue. The shoot base was fixed to the wall of the container with the root-shoot junction positioned 20 mm below the surface of the solution while the shoots remained in air. Radial O_2 loss appeared as blue halos around the roots, and photographs were taken 2 h after placing roots in the methylene blue solution. For plants grown in stagnant medium, root length, the length of the barrier to radial O_2 loss (i.e., colourless part of the roots), and the ratio of barrier to radial O_2 loss length to total root length was calculated.

2.6. Permeability of the outer part of the root to an apoplastic tracer

The permeability of the root apoplast was determined using the periodic acid-Schiff's reagent approach as detailed in Soukup et al. (2002). Periodic acid oxidizes cell walls, producing aldehydes that stain purple with Schiff's reagent. Twenty mm long root segments (for positions see below) were excised from 100–120 mm long adventitious roots and the cut ends were sealed with lanoline. The segments were then incubated in 0.1 % (w/v) H_5IO_6 for 1 h, washed with DI water and immersed for 1 h in a reducing solution containing 2 % (w/v) KI, 2 % (w/v) $\text{Na}_2\text{S}_2\text{O}_3$ and 2 % (v/v) 1 M HCl and subsequently rinsed with DI water. 100 μm cross-sections were made from the middle region of each root segment using a vibrating microtome (VT1200S, Leica Biosystems Nussloch GmbH, Germany), and the sections were stained with Schiff's reagent for 10 min and then immersed in 70 % (v/v) glycerol for observation. The purple colour, indicating periodic acid penetration into the cell walls, was visualized under a white light microscope (BX60, Olympus Optical Co. Ltd, Japan). In aerated and non-aerated solutions, the permeability of the root apoplast was assessed at 60 mm behind the root apex, whereas for plants grown in stagnant medium, it was assessed at 40, 50, 60, and 70 mm behind the root apex. Under aerated and non-aerated conditions methylene blue staining showed that O_2 leaked along the entire root length, and therefore the apoplastic tracer was not applied on more than one position behind the root apex. In contrast, O_2 leakage stopped about 30–60 mm behind the root tip under stagnant conditions, and therefore this zone was analysed in more detail for roots in stagnant medium (Fig. 3).

2.7. Apparent permeance to O_2 of the outer part of the root

We measured O_2 penetration into the root cortex and subsequently calculated the apparent O_2 permeance (P_A) of the outer cell layers according to the approach of Peralta Ogorek et al. (2021). Segments (20 mm long) were excised from 110 to 140 mm long adventitious roots at 50–70 mm behind the root apex from plants grown in aerated or non-aerated solutions and at 30–50 mm behind the root apex from plants grown in stagnant medium (barrier to ROL formed). Cut ends were sealed with lanolin to prevent longitudinal O_2 penetration. The root segments were carefully tied to a metal wire mesh with rubber bands and fixed in a 1-litre Perspex tank. Oxygen penetration was measured in the midpoint of each root segment with an O_2 microsensor (OX-25, Unisense A/S, Aarhus, Denmark) mounted on a motorized micromanipulator. A boom-stand stereo microscope was used to accurately position the microsensor on the root surface. The root segment was immersed in DI water with a pO_2 of 20.6 kPa (air-equilibrium), and an optical O_2 microsensor (OptoMR, Unisense A/S) was placed 5 mm away from the root segment to monitor the O_2 levels in the bulk water, along with a temperature sensor (ZNTC, Unisense A/S). The O_2 microsensor was then inserted vertically 175 μm into the root segment, ensuring it was within the cortex while avoiding penetration into the stele. O_2 partial pressures (pO_2) within the cortex and the nearby bulk water were recorded every 5 s with Unisense Logger (SensorSuite v 3.1, Unisense A/S). After a minimum of 10 min of steady-state data collection, the tank was emptied without retracting the O_2 sensor from the cortex and then refilled with DI water at a pO_2 of approximately 85 kPa. The sensor was then moved 10 μm further into the cortex to ensure tight sealing around the sensor before recording data for another 10 min. The apparent permeance to O_2 (P_A ; m s^{-1}) was calculated using the equation of Lendzian (2006)

$$P_A = \frac{F}{A \times \Delta C} \quad (2)$$

where F (mol s^{-1}) is the O_2 intrusion rate through the surface area A (m^2) of the root segment, and ΔC (mol m^{-3}) is the O_2 concentration gradient calculated as the O_2 concentration difference between cortex and bulk water. In cases where seedlings were grown in stagnant medium, roots developed a tight barrier to radial O_2 loss. Consequently, the pO_2 within the cortex remained at 0 kPa indicating that no O_2 diffused from the solution into the cortex.

2.8. The rate of radial water loss from root segments

Radial water loss from root segments was measured as described in Peralta Ogorek et al. (2021) and Song et al. (2023). Briefly, root segments without laterals were excised at 30–70 mm behind the root apex from 110–140 mm long adventitious roots, and the diameter of each segment was measured using a digital calliper. A minimum of 200 mg of fresh mass were prepared for each replicate, and these root segments were all obtained from a total of eight roots harvested from two plants within the same pot. The cut ends of the root segments were sealed with lanolin and maintained in a moist condition before any excess water was wiped off with paper towels before the root segments were placed on a 5-digit balance (Mettler Toledo Analytical Balance Xs105 with LabX direct balance 2.4 software) with desiccant (silica gel) in the weighing chamber to maintain low humidity ($< 30\%$, monitored by a relative humidity data logger, HOBO UX100–011, Onset). All measurements were conducted at 22–25 $^{\circ}\text{C}$. Cumulative water loss was automatically recorded every minute over 1 h after which the root segments were oven-dried at 50 $^{\circ}\text{C}$ to constant mass. Rates of radial water loss ($\mu\text{mol H}_2\text{O m}^{-2} \text{s}^{-1}$) were calculated as the loss in mass between two time points divided length of the time interval and divided by the surface area of the root segments. To facilitate comparisons between the different treatments, a specific time point was chosen, corresponding to when the root samples

had lost 15 % of their total water content; see Peralta Ogorek et al. (2021) for the rationale behind selecting this time point.

2.9. Histochemical staining of lignin and suberin in root cell walls

To visualize the cells associated with the apoplastic barriers of the outer part of the root, histochemical staining of lignin was conducted on root segments from stagnant medium without or with Si. 100 μm thick cross-sections at positions 40, 50, 60 and 70 mm behind the root apex of 100–130 mm long adventitious roots were collected from 5-week-old seedlings. Lignin depositions in root cross-sections were visualized using phloroglucinol-HCl (Mitra and Loqué, 2014) or Mäule reaction (Kutscha and Gray, 1972). Lignified cells appeared as red-pink (cinnamaldehyde group) in the former or brown-orange (syringil group) in latter method, while non-lignified tissues remained unstained under white light microscopy (BX60, Olympus Optical Co. Ltd, Japan). For the phloroglucinol reaction, cross-sections were treated with 2 % phloroglucinol (w/v in 96 % ethanol) for 3.5 min in darkness, then immersed into 20 % HCl for 1–2 min. For the Mäule reaction, cross-sections were treated with a 1 % (w/v) KMnO_4 solution for 30 s. Subsequently, the cross-sections were washed thoroughly with DI water and incubated in 12 % HCl before they were treated with 32 % ammonia. To facilitate the interpretation of lignification patterns of sclerenchyma cell wall, we established a lignification index to quantify intensity differences using ImageJ. Three straight lines (L_1 , L_2 and L_3 below) within the stained cross-sections (RGB images) were selected to extract colour value data (Fig. S3), and then the colour histogram function was used to obtain the mean RGB colour value (0 represents dark and 255 represents white). The mean RGB colour value represents the colour intensity associated with a specific area of the image, and a higher index value indicates a stronger lignification. The lignification index was calculated according to the following equation:

$$\text{Lignification index} = 300 - \frac{L_2 * 255}{(L_1 + L_3)/2} \quad (3)$$

where L_1 , L_2 and L_3 represent the mean RGB colour values of the three straight lines, respectively; L_2 is the target area (sclerenchyma) for analysis, L_1 and L_3 are adjacent cells representing benchmarks. The value 255 represents the mean RGB colour value of white and 300 is a unit-less constant.

Suberization of the cell walls of the outer part of the roots was visualized using Fluorol Yellow 088. 100 μm thick cross-sections were prepared as above and suberin lamellae were detected after staining with 0.01 % (w/v) Fluorol Yellow 088 as described in Brundrett et al. (1991). Suberin lamellae were visualized as a yellow-green fluorescence in cross-sections evaluated under UV excitation light (UV1-A filter, band pass = 365 nm, dichroic mirror = 400, barrier filter = 400) in a Nikon microscope (ECLIPSE Ci-L, Nikon) and photographed using a DS-Fi3 camera (Nikon, Tokyo, Japan).

2.10. Root tissue porosity and root anatomy

Root porosity was measured under stagnant conditions without or with Si. The tissue porosity of adventitious roots, expressed as the percentage of gas volume per unit of root volume, including both small intercellular spaces and aerenchyma, was assessed using the pycnometric displacement technique (Sojka, 1988). Six adventitious roots 100–130 mm in length and without any laterals, were selected from each pot. Subsequently, two 20 mm segments were then severed at distances of 30–50 mm and 50–70 mm behind the root apex, and the root tissue porosity was measured as described by Tong et al. (2023). The images of cross-sections from the root samples used for assessing permeability of the root apoplast were processed using ImageJ to quantify aerenchyma percentage, as well as root, cortex and stele diameters.

2.11. Data analyses

GraphPad Prism 8 (v.8.0.2, La Jolla, CA, USA) was used for graphs and statistical analyses. Mean comparisons were conducted via unpaired t-test or two-way ANOVA followed by a Sidák post hoc test. To test whether the datasets met the assumptions of the parametric analyses (normality and homoscedasticity), we examined the homoscedasticity and residual plots. Cortex to stele ratio data sets underwent log-transformation to meet the assumptions. Furthermore, the correlation between Na^+ and Cl^- fluxes was investigated using non-parametric Spearman-rank correlation coefficients, as bivariate data lacked normality. A non-linear correlation analysis between the Na^+ flux, Cl^- flux, and the lignification index was also conducted. All analyses used a significance level of $\alpha = 0.05$, with non-transformed data being presented in the figures. Further information about the number of replicates, *P*-values and the statistical test used are indicated in each figure caption.

To compare the effects of Si application on traits obtained under aerated, non-aerated and stagnant treatments, the Hedges' *g* index was calculated for each treatment as a measure of the standardized effect size (Del Re, 2012). Hedges' *g* is an unbiased estimate of Cohen's *d* not affected by sample size, scale of measure or magnitude. Null values indicate no effect of Si on traits, whereas positive or negative values imply an increase or decrease of the trait, respectively, caused by Si application. Effect sizes are considered negligible if lower than 0.2, of low impact when up to 0.5, of medium impact from 0.5 to 0.8, and of high impact if larger than 0.8 (Lakens, 2013).

3. Results

3.1. Na^+ and Cl^- concentrations in the xylem sap

The root barrier to radial O_2 loss has previously been shown to restrict Na^+ and Cl^- uptake in rice (Ranathunge et al., 2011). Here we showed that the Si application had a statistically significant impact on the barrier formation and function only under stagnant conditions, but not under aerated or non-aerated conditions (see below), and therefore we only assessed Na^+ and Cl^- intrusions under stagnant conditions. With Si, the sap volume flow was 2.8-fold higher compared to without Si with 0.1 MPa above atmospheric pressure applied to the root chamber (Fig. 1C). Since Si had no significant impact on root surface area (Fig. S4F), the increase in sap volume flow were likely caused by apoplastic bypass flow at the non-stress concentrations used and/or a higher transport capacity caused by the larger stele (Fig. S5C). Na^+ and Cl^- concentrations in the xylem sap were differentially affected by Si, resulting in a 1.9-fold increase in Na^+ but only a 1.3-fold increase in Cl^- (Fig. 1D, E). When converted into a flux (i.e., ion intrusion per root area per time unit), Si caused a significant 7.9- and 5.3-fold increase in the Na^+ and Cl^- fluxes, respectively (Fig. 1F, G). Regardless of Si application, the relationship between Na^+ and Cl^- fluxes was linear, but the flux of Cl^- was systematically higher than that of Na^+ as indicated by all of the data points positioned above the 1:1 line (Fig. 1H). The observed impact of Si on the fluxes of Na^+ and Cl^- can probably be attributed to effects of Si on bypass flow, and we therefore assessed the permeability of the outer apoplastic barriers.

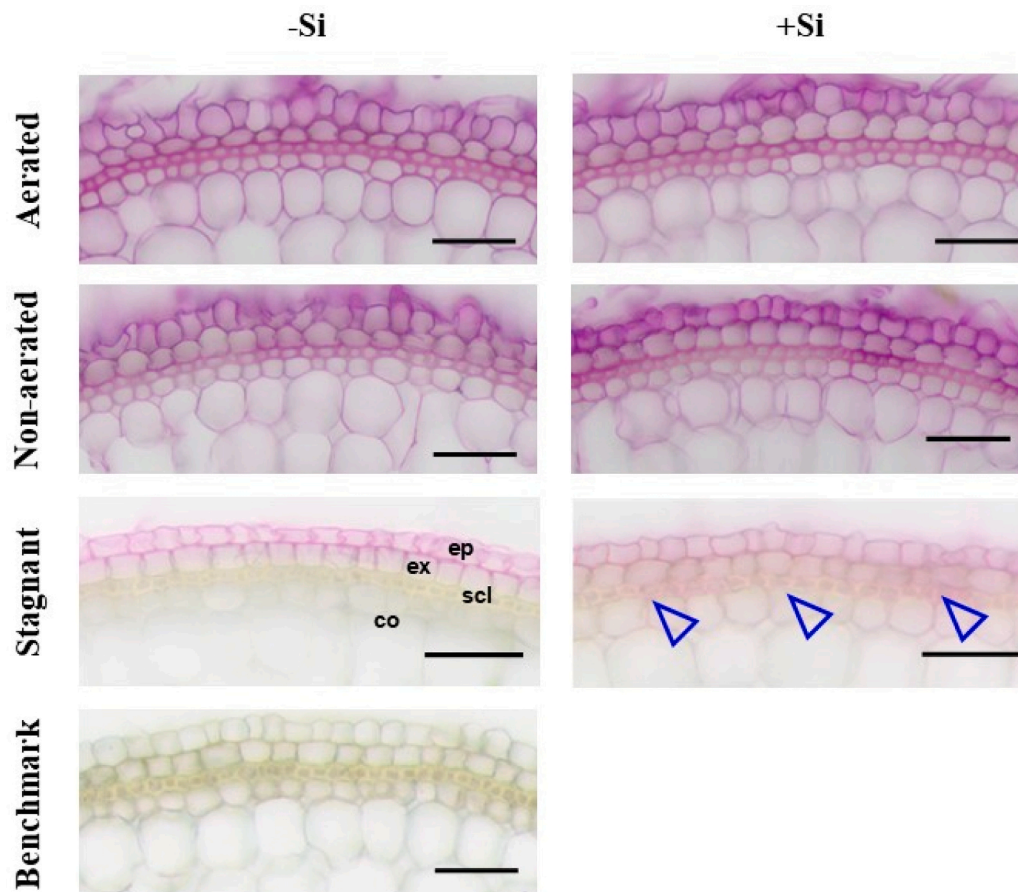


Fig. 2. Permeability of periodic acid in the outer part of the roots of *Oryza sativa* grown without or with 2 mM Si application. Cross-sections were taken at 60 mm behind the root apex of 100–120 mm-long adventitious roots formed under aerated, non-aerated and stagnant conditions without or with Si. Purple coloration indicates penetration of apoplastic tracer periodic acid into the root cortex. Blue triangles indicate partially stained cell walls under stagnant conditions with Si. A benchmark cross-section without staining is shown for comparison. Scale bar: 50 μm . Co, cortical cells; ep, epidermis; ex, exodermis; scl, sclerenchyma.

3.2. Permeability of adventitious roots

We used periodic acid as a tracer to visualize the permeability of the outer apoplastic barriers at 60 mm behind the root apex. In roots from aerated or non-aerated conditions, periodic acid penetrated the

exodermis and the sclerenchyma cells and reached the mid-cortex regardless of Si application (Fig. 2). However, under stagnant conditions and without Si, the tracer was only detected in the epidermal cell layers, indicating that the tracer was effectively blocked at the Casparian bands in the radial walls of exodermis. Under stagnant conditions with

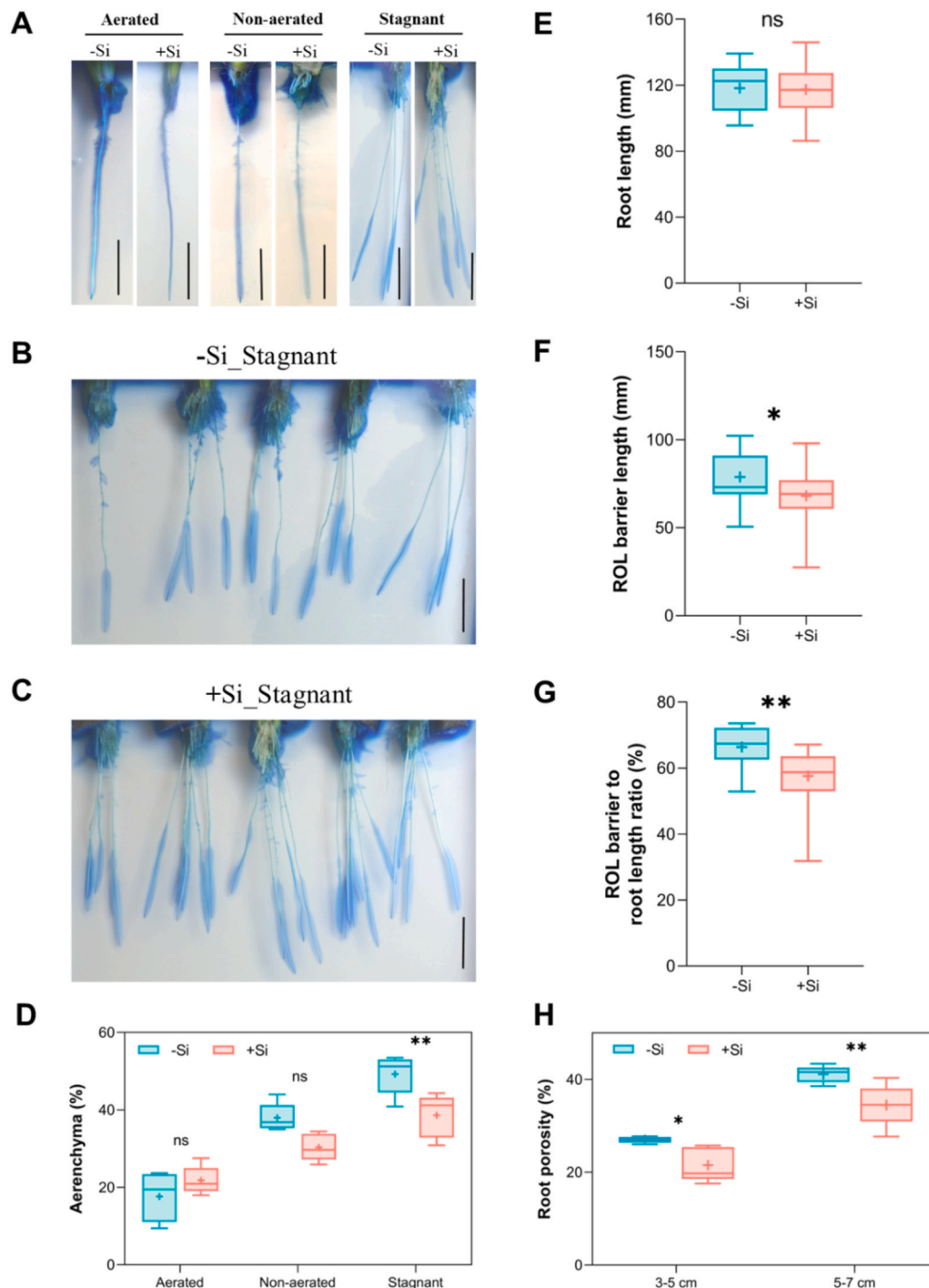


Fig. 3. Visualization radial O_2 loss from roots of *Oryza sativa* grown without or with 2 mM Si application. (A) Methylene blue staining indicates O_2 loss from roots grown in aerated, non-aerated or stagnant conditions without or with Si. Methylene blue staining indicates O_2 loss from roots grown in stagnant conditions without Si (B) or with Si (C). (E) Root length, (F) barrier to radial O_2 loss length, and (G) ROL-barrier-length:root-length ratio from plants grown without or with Si. (D) Aerenchyma percentage determined from the root cross sections at 60 mm behind the root apex of 100–120 mm-long adventitious roots grown under aerated, non-aerated or stagnant conditions without or with Si. (H) Root tissue porosity from plant grown in stagnant conditions without or with Si. In (D), two-way ANOVA showed significant 'Si', 'growing conditions' (nutrient solution aeration status) and 'Si x growing condition'; in (H), two-way ANOVA showed significant effects of 'distance' and 'Si', but no significant effect of 'distance x Si' (see Table S1 for P values and % of total variation). The box and whisker plot shows the mean (+) of five true replicates, the median (horizontal line), the first and third quartiles (box), and minimum and maximum (whiskers). Asterisks indicate significant differences (E–G, unpaired Students *t*-test; D, H, Šidák's comparisons test): **P* < 0.05, ***P* < 0.01, ****P* < 0.001; ns, not significant. Scale bars: 30 mm.

Si application, periodic acid penetrated into the cortex in a patchy manner suggesting that the formation of the apoplastic barrier in the exodermis was only partial (Fig. 2). A more detailed analysis conducted at multiple positions along the root grown under stagnant conditions, confirmed these observations conducted at 60 mm behind the root apex (see Fig. 7 and corresponding analysis).

3.3. Formation of a barrier to radial O_2 loss and development of a porous cortex

We used methylene blue staining to visualize the pattern of radial O_2 loss from roots as the dye forms blue haloes at the sites of O_2 loss. In both aerated and non-aerated nutrient solutions, irrespective of Si application, radial O_2 loss occurred along the entire root length (Fig. 3A). In contrast, in roots grown in stagnant conditions, blue haloes were observed only at the apical regions, whereas the basal regions appeared white indicating the presence of a strong barrier to radial O_2 loss (Fig. 3B, C). Without Si, the barrier to radial O_2 loss was formed at 39 ± 6 mm behind the root apex (Fig. 3B, F), whereas with Si, the barrier formation was significantly delayed and occurred at 49 ± 8 mm behind the root apex (Fig. 3C, F). Even though the root length was unaffected by Si treatment, it led to a 13 % decrease in the proportion of barrier to total root length since the barrier formation was delayed, highlighting the functional consequences of Si application on radial O_2 loss under stagnant conditions (Fig. 3E, G).

In addition to the barrier to radial O_2 loss, internal aeration in plant roots is also facilitated by aerenchyma and root tissue porosity (Armstrong, 1979). We therefore measured cortical aerenchyma at 60 mm behind the root apex for all treatments, i.e., $\pm$ Si under aerated, non-aerated and stagnant conditions. In neither aerated nor non-aerated conditions, Si application significantly affect cortical aerenchyma, which occupied 18–22 % of root cross-sectional area in aerated conditions and 30–38 % in non-aerated. In stagnant conditions however, lack of Si further increased aerenchyma significantly to 49 % (Fig. 3D).

Root porosity captures not only aerenchyma, but also gas-filled intercellular spaces and was measured in segments excised at 30–50 mm and 50–70 mm behind the apex in roots grown under stagnant conditions as there was a significant difference in the amount of aerenchyma in neither aerated nor non-aerated conditions (Fig. 3D). In root segments from position 30–50 mm, Si reduced root porosity significantly by 5 % (from 27 % to 22 %), and at position 50–70 mm the reduction was 7 % (from 41 % to 34 %) (Fig. 3H).

3.4. Apparent permeance to O_2 of the outer part of the root

We quantified apparent permeance to O_2 (P_A) of the outer part of the root to further characterize the influence of Si application on the permeability of the apoplastic barrier in the outer root cell layers. We diagnosed segments severed at 50–70 mm behind the root apex, where a barrier to radial O_2 loss would normally have formed when grown in stagnant conditions (Jiménez et al., 2021b). When the external pO_2 of the medium was maintained at 20.6 kPa (air-equilibrium), cortical pO_2 in segments without a barrier showed a slight initial decline due to O_2 consumption by the tissue, but it eventually reached a quasi steady-state (Fig. 4A). Subsequently, when the pO_2 of the bulk medium was raised to 85 kPa, the cortex pO_2 increased rapidly during the first minutes before stabilizing at a new quasi steady-state (Fig. 4A). However, the pO_2 inside the cortex never reached the same level as the external pO_2 due to O_2 consumption by respiration. In contrast, the segments with a tight barrier to radial O_2 loss showed a very different pattern than those without a barrier. The initial decline in pO_2 , when exposed to an air-equilibrium medium, continued until the cortex reached anoxia (detection limit), i.e., O_2 initially present in the cortex was respired. With 85 kPa external pO_2 , O_2 intrusion exceeding respiratory consumption was not observed (Fig. 4A).

Under neither aerated nor non-aerated conditions, Si application

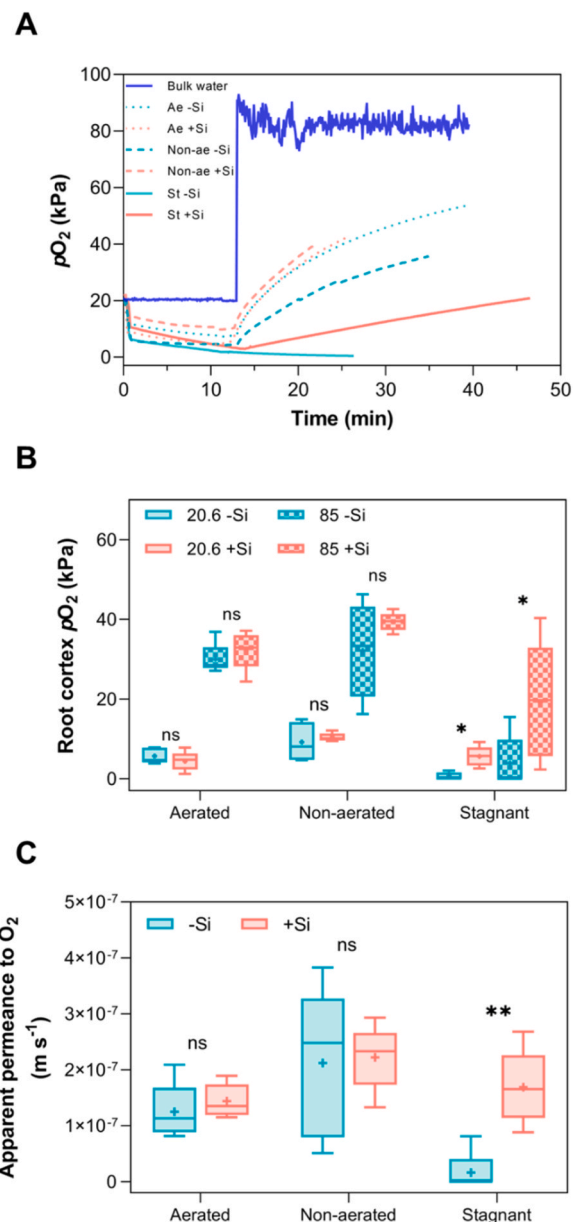


Fig. 4. O_2 intrusion in roots of *Oryza sativa* formed under aerated, non-aerated or stagnant conditions without or with 2 mM Si application. (A) Examples of time traces of O_2 intrusion into root segments from all treatments at two external pO_2 levels (external pO_2 changed from 20.6 kPa to 85 kPa after 13 minutes). (B) Steady-state cortex pO_2 values at the two external pO_2 levels, (C) the apparent permeance (P_A) to O_2 of the outer part of the root. Data were measured on 20 mm long segments (position 50–70 mm behind the root apex) from 110 to 140 mm-long adventitious roots formed under stagnant conditions. In (B), two-way ANOVA showed significant effects of ‘growing conditions’ and ‘growing conditions $\times$ Si’, but no significant effect of ‘Si’; in (C), two-way ANOVA showed significant effects of ‘growing conditions’ and ‘Si’, but no significant effect of ‘growing conditions $\times$ Si’ (see Table S1 for P values and % of total variation). The box and whisker plot shows the mean (+) of five true replicates, the median (horizontal line), the first and third quartiles (box), and minimum and maximum (whiskers). Asterisks indicate significant differences (Sidak’s comparisons test): * $P < 0.05$, ** $P < 0.01$; ns, not significant.

caused any significant difference in cortical pO_2 between the two external O_2 pressures applied (20.6 versus 85 kPa). However, under stagnant conditions with Si, cortical pO_2 was 8.8-fold at 20.6 kPa and 5.0-fold at 85 kPa higher compared to without Si (Fig. 4B). Since the estimation of P_A to O_2 is based on the intrusion rates, the P_A to O_2 also

showed no effect of Si application for aerated or non-aerated roots, whereas in stagnant conditions, Si caused a significant 10.4-fold increase in P_A to O_2 clearly demonstrating the effect of Si on the barrier to radial O_2 loss (Fig. 4C).

3.5. Radial water loss from roots

We quantified radial water loss to evaluate the effect of Si on the resistance to water flow across the outer part of the root. The highest cumulative water loss from root segments was found for the aerated and non-aerated treatments (regardless of Si treatment), which did not have a barrier to radial O_2 loss. These reached steady-state during the measurements, while those with a tight barrier (stagnant conditions without Si) showed a more linear response (Fig. 5A). Roots from stagnant conditions with Si exhibited an intermediate response, although they also followed a linear rate of water loss pattern similar to the root with a tight barrier. Regardless of Si, roots from either aerated or non-aerated conditions had lost about 80 % of their total water content within 30–45 min. Cumulative water loss was below 55 % even after 60 min for roots grown in stagnant conditions with Si treatment and only 28 % for roots from stagnant conditions without Si. According to Peralta Ogorek et al. (2021), approximately 15 % of the total water content within a rice root is bound in cells exteriorly the exodermis, and thus we conducted a more detailed analysis around the timepoint where 15 % of the water content had been lost. In both aerated and non-aerated roots, 15 % was lost within 5 min, in contrast to 10 and 25 min for stagnant condition with and without Si, respectively (Fig. 5A). After 60 min, rates of radial water loss have stabilized at $300 \mu\text{mol m}^{-2} \text{s}^{-1}$ for all treatments, except for the roots from stagnant conditions with Si showing a rate of $700 \mu\text{mol m}^{-2} \text{s}^{-1}$ as these roots still contained lots of water (Fig. 5B). Rates of radial water loss are driven by the humidity gradient between tissue and air, and to facilitate comparison between different treatments, we calculated the rates at the time point when the roots had lost 15 % of their initial water content. These comparisons showed that Si did not affect radial water loss in either aerated or non-aerated conditions. However, in stagnant conditions, Si caused a significant 3-fold loss of water, demonstrating the considerable effect of the barrier to radial O_2 loss in also restricting water loss (Fig. 5C).

3.6. Histochemical staining of lignin and suberin in the outer parts of the root

We analysed lignification of roots from the stagnant medium using both a qualitative and a semi-quantitative approach as lignification of the sclerenchyma cell walls is supposedly one of the major components of the outer apoplastic barriers (Peralta Ogorek et al., 2023). Lignin deposits were visualized in root cross-sections by red/pink staining during the phloroglucinol reaction or by brown staining during the Mäule reaction. In the outer cell layers, lignin deposition was mainly observed in the sclerenchyma layers and anticlinal exodermal cell walls in roots from stagnant conditions, regardless of Si application (Fig. 6). The degree of lignin deposition increased along the root from the apex to the base. At 40 mm behind the root apex, weak staining was observed in the outer root cell layers, whereas at 70 mm, strong staining occurred in both exodermis and sclerenchyma cell walls. Some cortical cells immediately inner to the sclerenchyma layer had thick, lignified walls (cortical sclerenchyma). Furthermore, Si treatment significantly affected the lignin deposition in sclerenchyma cell wall in which the colour intensity at similar positions was weaker compared to those without Si treatment (Fig. 6). To better illustrate the effect of Si on lignin deposition, we established a lignification index, where a high numerical value indicates higher colour density. Using this semi-quantitative approach for both staining procedures, we showed that Si significantly reduced or delayed the lignification in all positions along the roots (Fig. 6), except for the phloroglucinol reaction at position 60 mm (Fig. S2D).

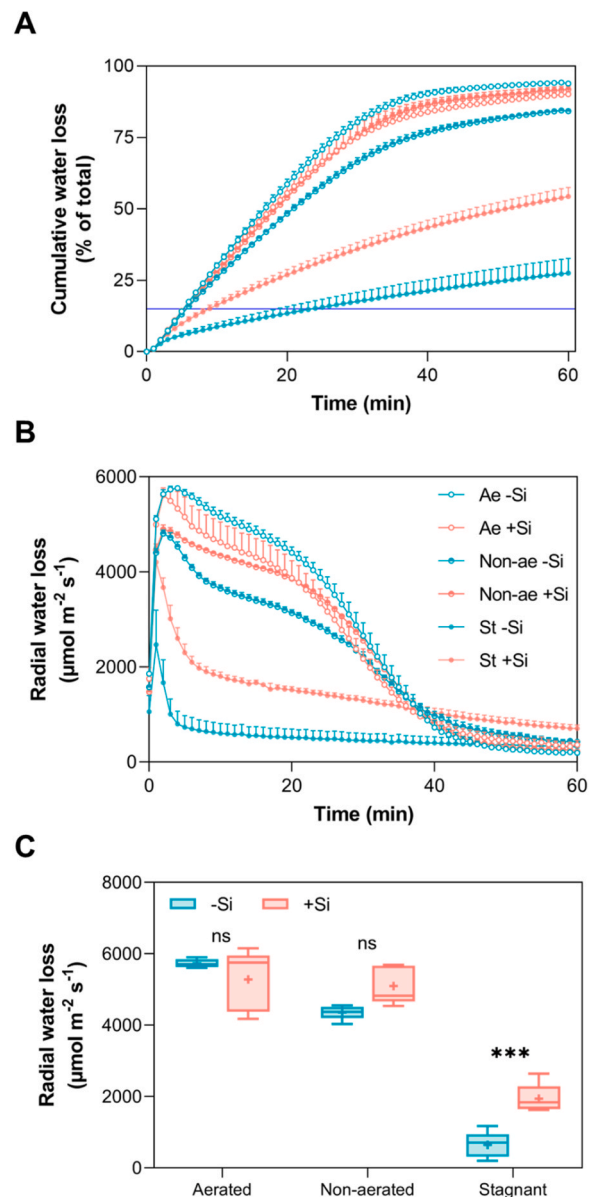


Fig. 5. Radial water loss (RWL) from roots of *Oryza sativa* grown under aerated, non-aerated or stagnant conditions without or with 2 mM Si application. (A) Percent cumulative water loss (CWL) from the total water content of root segments plotted against time and (B) RWL from roots under aerated, non-aerated or stagnant conditions without or with Si. Data are from 40 mm long root segments exposed to dry air (RH = 20–30 %) taken from 30 mm behind the root apex of 110–140 mm long adventitious roots. In (A), the blue horizontal line indicates 15 % of CWL, and the RWL data in (B) are presented at the point when 15 % of tissue water had evaporated. In (C), two-way ANOVA showed significant effects of ‘growing conditions’, ‘Si’, and ‘growing conditions × Si’ (see Table S1 for P values and % of total variation). The box and whisker plot shows the mean (+) of five true replicates, the median (horizontal line), the first and third quartiles (box), and minimum and maximum (whiskers). Asterisks indicate significant differences (Šidák’s comparisons test): *** $P < 0.001$; ns, not significant.

Under adverse environmental conditions (e.g., abiotic stresses), suberin can also be deposited in the cell walls of the outer parts of roots, and it is considered an import component of outer apoplastic barriers including the barrier to radial O_2 loss (Shiono et al., 2014). Unlike the lignification of sclerenchyma cell walls, suberization mainly takes place in the cell walls of the exodermis, which is evident from our histochemical staining for suberin (Fig. 7). As for suberin, when Si was added,

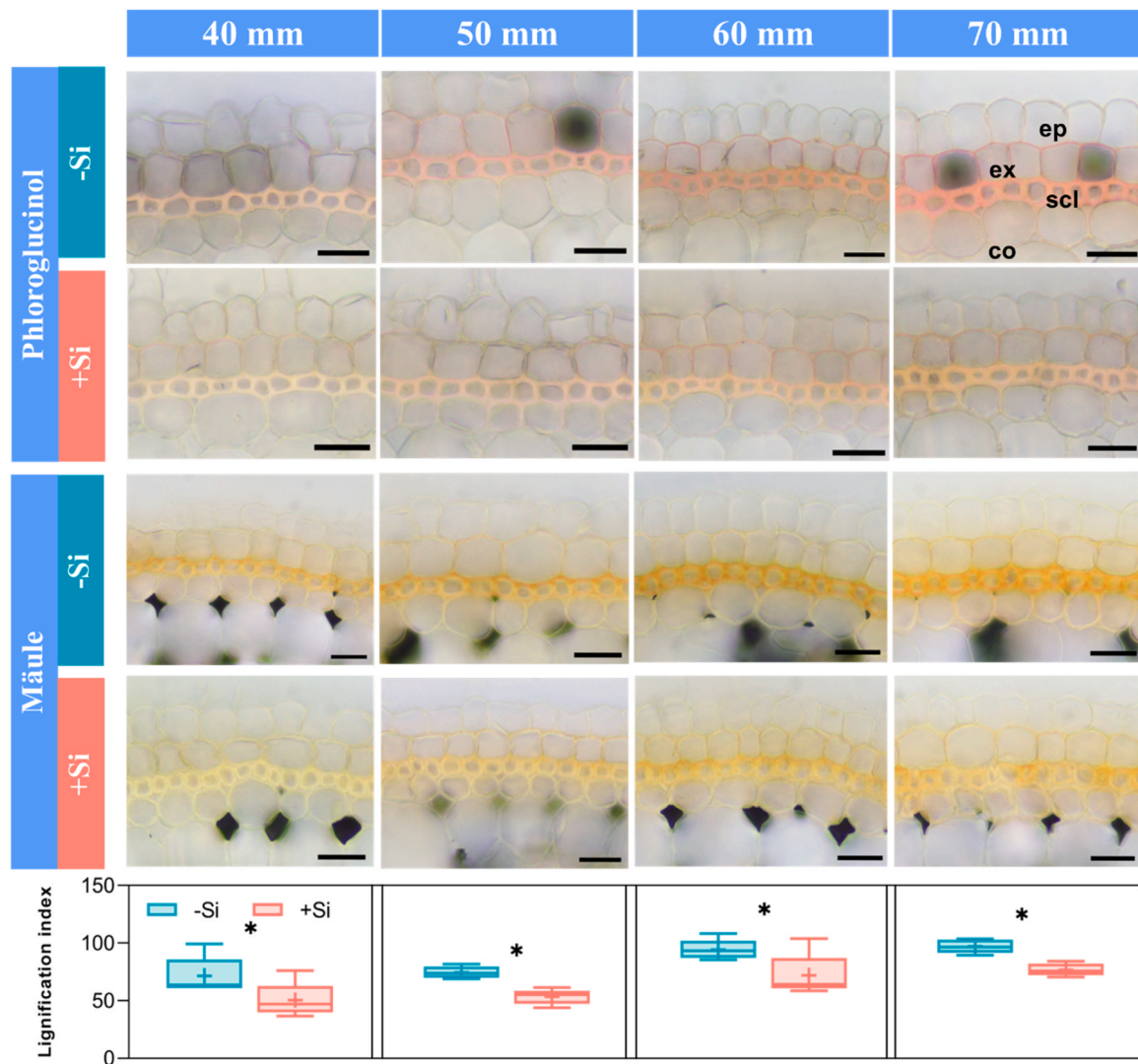


Fig. 6. Histochemical staining and quantification of lignin deposition in the outer part of the roots of *Oryza sativa* grown in the stagnant medium without or with 2 mM Si application. Cross-sections were taken at the positions of 40, 50, 60 and 70 mm behind the apex of 110–140 mm long adventitious roots grown without or with Si. Cinnamaldehyde groups of lignin in the cell walls appear as redpink staining with the phloroglucinol reaction, while syringil lignin groups appear as a brown-orange staining using the Mäule reaction. Co, cortical cells; ep, epidermis; ex, exodermis; scl, sclerenchyma. Scale bars: 20 μ m. The lower panels show the lignification index for the outer part of the roots as based on the Mäule reaction. Two-way ANOVA showed significant effects of ‘distance’ and ‘Si’, but no significant effect of ‘distance $\times$ Si’ (see Table S1 for P values and % of total variation). The box and whisker plot shows the mean (+) of five true replicates, the median (horizontal line), the first and third quartiles (box), and minimum and maximum (whiskers). Asterisks indicate significant differences (Šidák’s comparisons test): * $P < 0.05$, ** $P < 0.01$; ns, not significant.

suberin deposition was very scarce at positions 40 and 50 mm behind the root apex, whereas suberization was already pronounced at these positions without Si. Although suberization occurred at positions 60 and 70 mm behind the root apex of Si-treated plants, the staining intensity was still lower compared with those positions of roots in the absence of Si (Fig. 7).

In accordance with the observed weak deposition of lignin and suberin in the outer part of the root, the apoplastic tracer (periodic acid) movement into the cortex was more pronounced in the roots from Si treatment compared with the roots without Si treatment (Fig. 7). In conclusion, our findings demonstrate a progressive increase in lignin and suberin depositions in the outer cell layers along the roots, from the apex to the base. Furthermore, Si application leads to a reduction in lignin and suberin deposition with implications for the tightness of the outer apoplastic barriers.

3.7. Rice growth and root anatomy

The distinct responses of rice to Si application under different treatments emphasize the role of the growth environment in modulating the effects of Si. We found that Si application reduced the number of roots by 21 % in stagnant and 37 % in non-aerated treatments, as compared to their respective counterparts without Si (Fig. S4D). In stagnant conditions, Si caused a 18 % reduction of maximum of root length (Fig. S4E). In non-aerated conditions, Si application caused a substantial decrease of 62 % of root DM, while there was no effect of Si in aerated or stagnant conditions (Fig. S4A). Interestingly, shoot DM increased 1.5-fold with Si application in aerated condition (Fig. S4B), highlighting the growth-promoting role of Si in certain environments (Fig. S4C). Under stagnant conditions, the cortex to stele ratio (Fig. S5D) was unaffected by Si, whereas root diameter increased 1.2-fold, cortex thickness 1.2-fold, and stele diameter 1.1-fold (Fig. S5A–C). No significant effects of Si application were found for these traits in plants grown in aerated or non-aerated conditions.

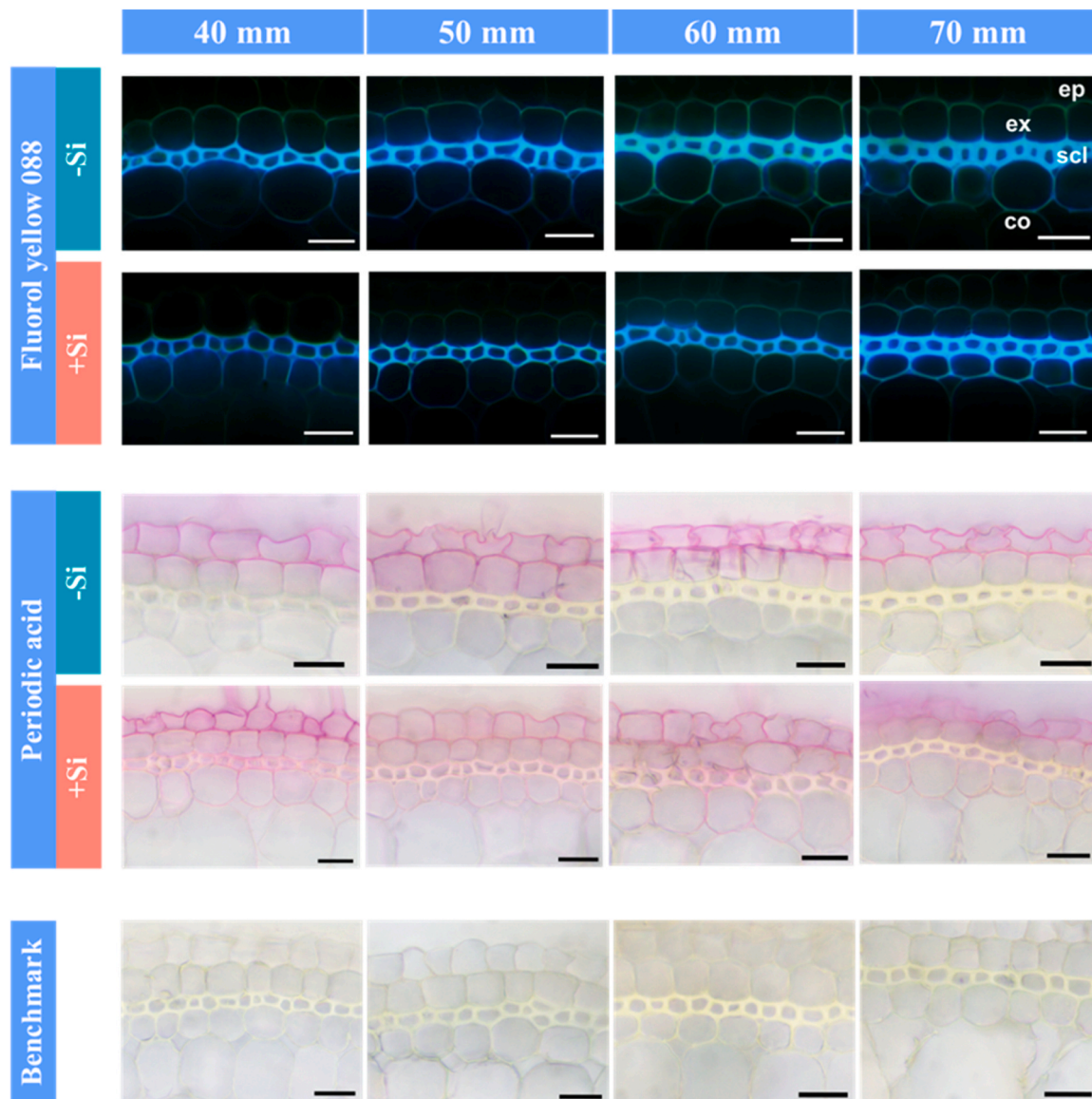


Fig. 7. Histochemical staining of suberin depositions, and penetration of an apoplastic tracer in the outer part of the roots of *Oryza sativa* grown in the stagnant medium without or with 2 mM Si application. The upper panels show typical cross-sections taken at positions at 40, 50, 60 and 70 mm behind the apex of 110–140 mm long adventitious roots grown without or with Si and stained with Fluorol Yellow 088; suberin lamellae are visible as yellow/greenish fluorescence in the exodermis. All replicate cross-sections are available at <https://doi.org/10.17894/ucph.e817dd86-4ec2-41ed-8a5b-4d267d76e03b>. The lower panels show the movement of purple apoplastic tracer (periodic acid) through the cell walls in cross-sections taken at the same positions with unstained benchmark cross-sections included for reference. Scale bars: 20 μ m. Co = cortex, ex = exodermis, scl = sclerenchyma and ep = epidermis.

3.8. Visualization of Si effects on multiple traits

The forest plot summarizes the multiple effects that Si application had on rice anatomy, morphology, physiology and growth. Si application had a large negative impact on plant growth in non-aerated conditions (e.g., shoot and root biomass, number of roots), also resulted in Si-driven inhibition of root acclimation to soil flooding. Interestingly, Si had positive effects on shoots in plants grown under aerated and stagnant conditions (Fig. 8A). Impacts on roots were negligible in the case of aerated conditions but potentially harmful under stagnant conditions. Here, roots showed a decrease in cortical aerenchyma and in P_A to O_2 . Although with Si, roots with less aerenchyma are expected to suffer from tissue O_2 deficiency, internal aeration can be compensated by the development of fewer and shorter roots (Fig. 8A). In fact, the cortex to stele ratio was remarkably lower for the non-aerated treatment compared to either the aerated or the stagnant treatments.

In stagnant conditions, higher P_A to O_2 and RWL strongly reflects the

weakening of the barrier to radial O_2 loss in presence of Si. This evidence is also confirmed by the remarkably lower lignification index in sclerenchyma cell walls observed along the entire root length after Si application (Fig. 8B). The weakening of the barrier to radial O_2 loss in presence of Si seems to allow larger ion entry into roots, as shown by the higher Na^+ and Cl^- concentrations in the xylem sap (Fig. 8B). Importantly, there was a significant negative correlation between the lignification index and the observed Na^+ and Cl^- concentrations in the xylem (Fig. 8C), as a low numerical value of the lignification index was reflected in an increase in Na^+ and Cl^- intrusion. Moreover, in presence of Si, the increase of ion fluxes in roots was exponential, suggesting a threshold in the amounts of lignin deposited in the root cell walls necessary to restrict Na^+ and Cl^- intrusion and to protect the plant from salt stress.

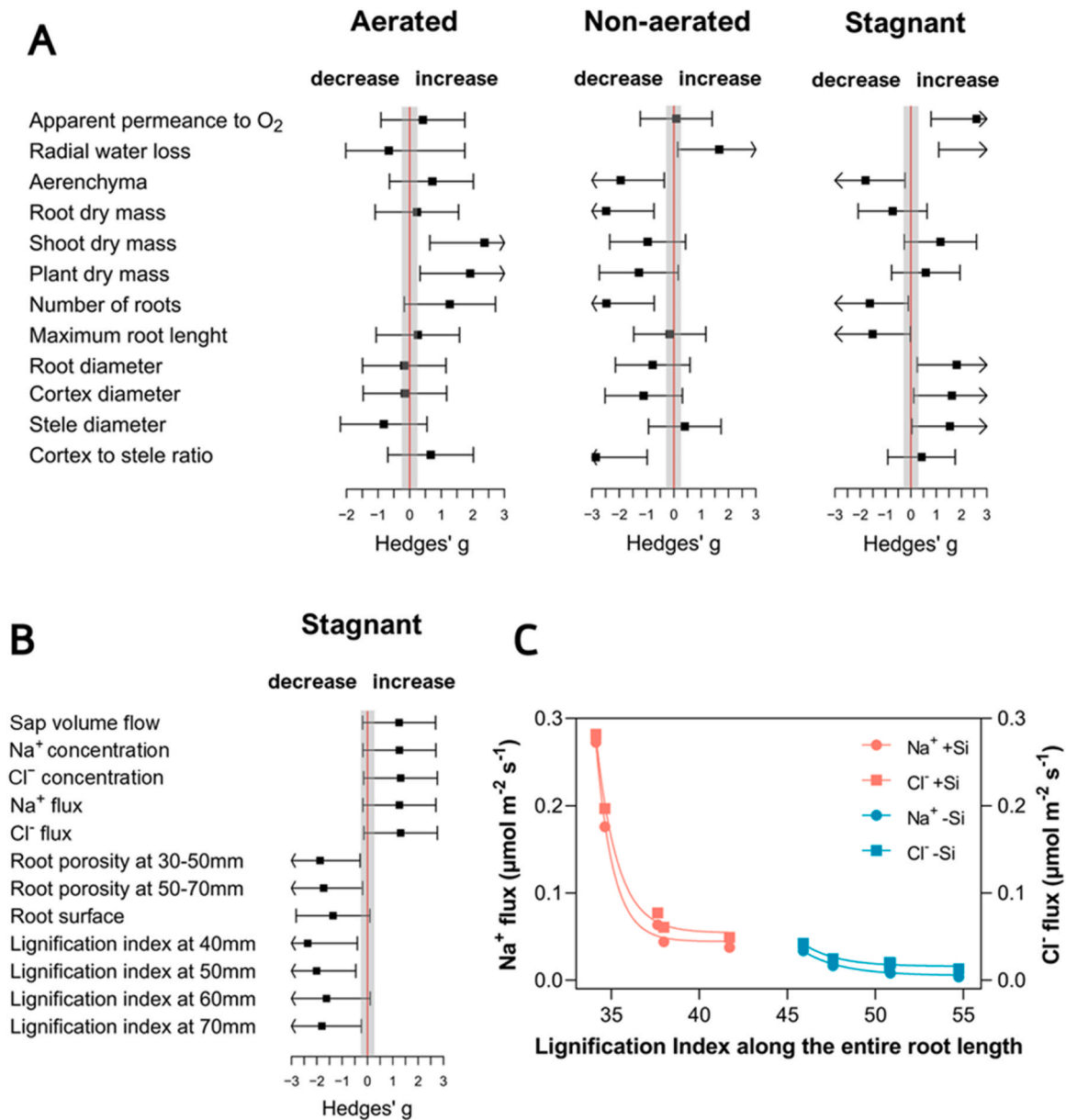


Fig. 8. The effect of Si on key root traits of *Oryza sativa* grown under aerated, non-aerated or stagnant deoxygenated conditions. Strengthening (positive values) or weakening (negative values) of traits in presence of Si are shown using forest plots of mean effect sizes, based on Hedges' g index (A, B). Effect sizes (g) lower than 0.2 are considered not significant and displayed within a grey band, and bars represent 95 % confidence intervals. Consequences for root lignification (phloroglucinol reaction) and ion fluxes were further investigated under stagnant deoxygenated conditions without and with Si and presented as a forest plot (B) and the relationship between the lignification index (distance behind from root apex) and ion fluxes (C).

4. Discussion

We aimed at assessing the impacts of Si on key root traits known to confer tolerance to soil flooding in rice, using hydroponics to mimic drained or flooded soils. We found that Si application had no effect on the formation of constitutive aerenchyma when the plants were grown in aerated medium (mimicking an O₂-rich drained soil) or inducible aerenchyma in non-aerated medium (mimicking the early stages of soil flooding with hypoxia). However in stagnant medium (mimicking advanced stages of soil flooding), Si caused lower amounts of inducible cortical aerenchyma, and weakened the barrier to radial O₂ loss and delayed its formation. The latter was reflected by weak lignin and suberin staining in the outer cell layers of the roots. The weak barrier was reflected in the higher concentrations of Na⁺ and Cl⁻ in the xylem sap when the roots were exposed to 100 mM NaCl in the Scholander

pressure bomb. Below we discuss these findings in the context of existing knowledge, and we provide recommendations on Si application as a tool to alleviate abiotic stress in an agricultural context highlighting the translational value of the present study.

4.1. Si weakened the barrier to radial O₂ loss and delayed its formation

The impact of Si on the development and structure of apoplastic barriers in roots has remained unresolved. Their composition and formation have been reported to be enhanced, unaffected, or even delayed, and different responses relates to both root age and species (Kreszies et al., 2020). However, in the impressive compilation of literature data, important information related to chemical or physical conditions of the hydroponic cultures are missing, i.e., there is no notion of whether the nutrient solutions were aerated, non-aerated, or stagnant, and there is

also no information on the O_2 status of the solution (Kreszies et al., 2020). We therefore designed our study to fill this knowledge gap by examining the effects of Si on apoplastic barrier formation and other root anatomical traits that are known to influence soil flooding tolerance in aerated, non-aerated and stagnant medium to cover the whole range of O_2 status from well aerated to effectively zero.

Roots of cultivated rice can form an inducible barrier to radial O_2 loss when grown in a stagnant medium (Fig. 3A, (Colmer, 2003b; Colmer et al., 1998). The barrier primarily locates in the mature basal parts (i.e., root zones more than 30–50 mm behind the root apex) whereas in younger developing zones (i.e., apex to 30 mm) there is no barrier formed or it is in the early developmental patchy stages (Colmer, 2003a, b; Jiménez et al., 2021b). Using methylene blue staining, we demonstrated the areas of radial O_2 loss from roots. We found that under stagnant conditions, the barrier had formed about 40 mm behind the root apex without Si application, whereas with Si it formed further back (about 50 mm behind the root apex (Fig. 3B, C). Since the adventitious roots had the same extension rate both without and with Si, we are concluding that Si application led to a delay in the formation of the barrier to radial O_2 loss in the outer part of the root.

In addition to the delayed formation of the barrier to radial O_2 loss caused by Si application, we observed that once formed, the barrier was also weaker than when Si was applied as indicated by higher permeance to O_2 . The permeability of the barrier was assessed by measuring the apparent permeance (P_A) of the outer part of the root to O_2 , and these measurements clearly showed a weaker barrier in the presence of Si compared to without Si; P_A was 6.4-fold higher with Si (Fig. 4C). The P_A to O_2 is a relatively new trait introduced in soil flooding tolerance, but the same genotype (IR42) as used in the present study has previously been diagnosed using an intermediate (0.1 mM) Si concentration. Comparing the P_A to O_2 at the three Si concentrations [(0 (this study), 0.1 (Peralta Ogorek et al., 2021; Tong et al., 2023) and 2.0 mM (this study)] does not reveal a straightforward dose-dependent response mostly because the P_A to O_2 was below detection limit in Peralta Ogorek et al. (2021). However, without Si the apparent permeance of the outer part of the root to O_2 was similar to the rate in Tong et al. (2023) using 0.1 mM and only when 2 mM was applied, it resulted in a significantly higher P_A to O_2 .

The weaker outer apoplastic root barriers were also reflected in the measurements of radial water loss (RWL). With Si application, the rates of RWL were 3.9- to 5.5-fold higher in the present study compared with 0.1 mM Si treatment (Peralta Ogorek et al., 2021; Tong et al., 2023). Moreover, the rates of RWL were still about 2-fold higher in the studies of Peralta Ogorek et al. (2021) and Tong et al. (2023) when compared with 0 mM Si in the present study. Hence, the rates of RWL show a more straightforward dose-dependent response as compared to P_A to O_2 .

Suberin depositions in the exodermal cell walls have been suggested to be the primary constituents of the barrier to radial O_2 loss (Colmer et al., 2019; Peralta Ogorek et al., 2024; Shiono et al., 2014). However, recent work demonstrates that low concentrations of H_2S primarily induces the formation of a barrier to radial O_2 loss through upregulations of genes associated with lignin rather than suberin biosynthesis (Peralta Ogorek et al., 2023). Moreover, barrier induction of low molecular carboxylic acids also showed upregulation of lignin-related genes (Colmer et al., 2019). In this study, the qualitative lignin assessment using histochemical staining indicates that Si application leads to a reduction in lignin deposition (Fig. 6) contradicting previous research where Si enhanced lignification in roots of rice (Fleck et al., 2011). The discrepancy could be attributed to i) different cultivars used, or ii) contrasting growing conditions, where rice was cultivated under non-aerated conditions in Fleck et al. (2011) instead of stagnant conditions (present study). The present study also supports the more common notion that suberin is the main functional component of the barrier to radial O_2 loss in rice (Shiono et al., 2014). Although the pattern was not as clear as for lignin, suberization was more pronounced without Si (with a stronger barrier to radial O_2 loss indicated by low P_A to O_2). Even

at positions 60–70 mm behind the root apex, suberin depositions in the cell walls of the exodermis was more pronounced without Si compared with Si applied (Fig. 7). However in the present study, we are unable to nominate one of the two key polymers are more important than the other for the barrier restricting O_2 diffusion.

4.2. Si application led to increased Na^+ and Cl^- in the xylem sap

Na^+ and Cl^- can move from the soil to the root xylem via two contrasting routes, i.e., by following the apoplastic pathway or through the cell-to-cell pathway, which is the combination of both symplastic (through plasmodesmata) and transmembrane (through plasma membrane transporters) paths. At the non-stress concentrations used, the Si-induced increase in Na^+ and Cl^- concentrations in the xylem sap can thus be due to a combination of increased bypass flow via the apoplast and increased transport via the cell-to-cell path. However, the roots of rice are known to be leaky and therefore the apoplastic pathway is considered quantitatively more important than the symplastic pathway for salt movements (Gong et al., 2006; Ranathunge et al., 2004). The bypass mechanism allows ions to enter the xylem through the immature endo- and exodermis or via emerging laterals, without crossing cell membranes or through passage cells in the endo- and exodermis of mature root zones, which do not possess suberin lamellae (Ranathunge et al., 2005). Si application has previously been shown to mitigate salt stress in rice by blocking the apoplastic bypass flow of Na^+ from root to shoot (Gong et al., 2006; Yeo et al., 1999), but we found that Si increased Na^+ and Cl^- concentrations in the xylem sap (Fig. 1F, G). Previous studies involving the impact of Si on bypass flow of Na^+ flux and Cl^- were conducted using aerated or non-aerated conditions (Gong et al., 2006; Yan et al., 2021; Yeo et al., 1999). However, in the present study, Si application only affected the outer apoplastic barriers when in stagnant medium; in aerated and non-aerated conditions, the barriers remained unaffected (Figs. 2–5). This is also the reason why we only conducted analysis of Na^+ and Cl^- concentrations in the xylem sap under stagnant conditions. Importantly, the finding that Si application increased bypass flow of Na^+ flux and Cl^- , weakened the barrier to radial O_2 loss, and delayed its formation might be caused by the different growing conditions in stagnant medium.

Furthermore, we found a strong correlation between Na^+ flux and Cl^- flux (Fig. 1F, G, H), suggesting that Si application had a similar influence on both Na^+ and Cl^- transport. However, the exclusion of Na^+ was systematically higher than that of Cl^- as indicated by all of the Cl^- data point positioned above the 1:1 line in Fig. 1H. The ability to restrict Na^+ intrusion more than Cl^- seem to be a common trait of rice regardless if the cultivars tested are salt tolerant or sensitive (Kumar and Khare, 2016). Nevertheless, the relationship between Na^+ and Cl^- and the lignification index assessed for the outer part of the root strongly suggests that the increase in Na^+ and Cl^- is due to bypass flow, a conclusion supported by a previous study reporting a similar relationship between suberin deposition in the cell wall of the outer part of the root and Na^+ bypass flow (Krishnamurthy et al., 2011).

4.3. Si application impacted key root traits involved in soil flooding tolerance

Si application not only affected the formation of apoplastic barriers in the outer part of the root that are involved in restricting radial O_2 loss and protection of roots from soil phytotoxins (Liu and Kreszies, 2023; Peralta Ogorek et al., 2023) but also other well-known root structural traits that are involved in soil flooding tolerance. Of these well-known traits, cortical aerenchyma is likely the most prominent as it enables diffusion of molecular O_2 over long distances from shoot to root apices due to the low diffusive resistance in the gas phase (Armstrong, 1979). Constitutive aerenchyma was formed in the cortex of roots grown in aerated nutrient solution, and the amount of ~20 % was unaffected by Si application (Fig. 3D). Once exposed to non-aerated or stagnant

conditions, inducible aerenchyma increased root tissue porosity to >30 % but the effect of Si was only statistically significant in stagnant conditions, where Si application caused a reduction in inducible aerenchyma (Fig. 3D, E). The observed responses of inducible aerenchyma in non-aerated or stagnant conditions are in accordance with existing knowledge within the field (Colmer, 2003a), whereas to our best knowledge, Si-induced effects on root cortical aerenchyma formation have not previously been reported for rice or any other crops.

Several other root traits as well as some shoot-related traits responded to the combination of growing conditions, i.e., aerated, non-aerated or stagnant medium, and Si application. All these responses are summarized in Fig. 8A; for example, the root diameter, cortex diameter and stele diameter increased with Si application, but only in stagnant conditions, whereas root mass, shoot mass and the total plant mass only increased in aerated conditions as a response to Si application. Thus, the otherwise well-establish growth promotion by Si (Epstein, 1999) was not consistent in the present study likely because there was no stresses imposed to the plants other than the simulated soil flooding.

For the additional plant response parameters obtained only for plants grown in stagnant medium, the summarizing picture provided by the forest plot is convincing. It clearly shows that sap volume fluxes, concentrations of Na^+ and Cl^- in the xylem sap and fluxes of Na^+ and Cl^- or the apoplastic bypass flows increased as a response to Si application, whereas lignification of sclerenchyma cell walls and root porosity declined (Fig. 8B). Therefore, even if shoot growth is enhanced by Si, long-term effects of the treatment could be unfavourable for the overall plant tolerance to abiotic stresses (flooding and salinity in particular), although this needs testing under real-world growth conditions in soils.

4.4. Conclusions

Under stagnant growing conditions, but not under aerated or non-aerated conditions, Si application weakened or delayed the development of key root traits associated with soil flooding tolerance. In stagnant conditions, Si application resulted in developing a weaker barrier to radial O_2 loss in the outer part of the root, and the barrier was formed further behind the root apex resulting in higher apparent permeance to O_2 and higher rates of radial water loss compared to roots grown without Si. The Si application also increased transport of Na^+ and Cl^- from the medium to the root xylem, reflected by greater bypass flows, and then higher concentrations of both ions in the xylem sap compared to plants without Si. The higher Na^+ and Cl^- bypass flow, the higher apparent permeance to O_2 and the higher rates of radial water loss were negatively correlated with lower depositions of lignin in sclerenchyma and suberin in exodermis cell walls indicating lower resistance in the outer part of the root to gas diffusion, and water and NaCl transport due to the formation of weak apoplastic barriers. Moreover in stagnant conditions, Si application significantly reduced inducible aerenchyma in the roots, whereas the adventitious roots became thicker due to a thicker stele and a thicker cortex. Consequently, we do not recommend using Si application to alleviate a combination stress of salinity and soil flooding in rice, until the present findings have been confirmed in soils since the barrier to radial O_2 loss and aerenchyma formation were both less pronounced in roots with Si application.

List of abbreviations

DI: Deionized; P_A : apparent permeance; $p\text{O}_2$, O_2 partial pressures; RH, relative humidity; ROL: Radial O_2 loss; RWL: Radial water loss; Si: Silicon.

Ethics approval and consent to participate

Not applicable.

Consent for Publication

Not applicable.

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CRediT authorship contribution statement

Kosala Ranathunge: Writing – review & editing, Conceptualization. **Elisa Pellegrini:** Writing – review & editing, Writing – original draft, Data curation. **Shuai Tong:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Juan de la Cruz Jiménez:** Writing – review & editing, Writing – original draft. **Dan Liu:** Data curation. **Max Herzog:** Conceptualization. **Lukasz Kotula:** Writing – review & editing, Conceptualization. **Ole Pedersen:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Availability of data and material

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request. Replicate root cross-sections for suberin staining are all available at a data repository, <https://doi.org/10.17894/ucph.e817dd86-4ec2-41ed-8a5b-4d267d76e03b>.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.envexpbot.2024.105941](https://doi.org/10.1016/j.envexpbot.2024.105941).

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